

AERIS: Boundary-Mapped Reliable Routing for Wireless Sensor Networks under Heterogeneous Channels

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Paper under double-blind review

Abstract—AERIS is a rule-based hierarchical routing protocol for wireless sensor networks under heterogeneous channels. We evaluate it with a corrected NS-3 evidence chain, including a seven-protocol dual-machine sweep against LEACH, PEGASIS, HEED, TEEN, CTP, and RPL-MRHOF across four environments and 50–1000 nodes ($n = 30$ per environment-node-protocol cell). The expanded audit changes the claim from a global ranking to a boundary map. Against the classical WSN baselines, AERIS is rank-1 in 21 of 28 environment-node cells and top-2 in all 28. After adding CTP and RPL-MRHOF, AERIS remains rank-1 mainly in outdoor suburban cells (6 of 7 scales, with a near tie at 1000 nodes), while CTP leads benign office cells and RPL-MRHOF leads most factory and urban cells. A new 3360-run NS-3 ablation and a corrected 400-replicate mechanism study show that AERIS’s gain is carried mainly by Gateway-assisted uplinks, while first-node death falls to 1–4.5 rounds in most harsh 500/1000-node cells. The result is a sharper deployment rule: AERIS is useful when simple, auditable WSN routing needs stronger delivery than classical baselines provide, but it is not a replacement for richer collection-tree or RPL-style stacks in all regimes.

Index Terms—wireless sensor networks, reliable routing, cross-platform evaluation, deployment boundary

I. INTRODUCTION

Wireless sensor network (WSN) routing is usually discussed in terms of energy, yet in many deployments link stability matters just as much [1]–[3]. This becomes clear when classical protocols are placed side by side: LEACH emphasizes clustered forwarding [4], PEGASIS uses chain-based aggregation [5], HEED relies on residual-energy-aware head election [6], and TEEN reports only when event thresholds are crossed [7]. Real links make the trade-off harder because low-power WSN channels are asymmetric, environment sensitive, and often unstable near connectivity thresholds [8]–[10].

This leads to a practical question: when do simple, auditable control rules outperform classical energy-first designs under harsh channels, and when are richer collection-tree or RPL-style stacks the better answer? AERIS addresses the first half of this question with fixed-rule control rather than learned routing. Its design combines Context-Adaptive Switching (CAS) for intra-cluster mode selection, Gateway reinforcement for CH-to-BS uplinks, and a sparse Skeleton fallback backbone.

We make three contributions. First, we present AERIS as a fixed-rule routing design whose online behavior can be traced directly to channel and topology inputs. Second, we evaluate AERIS with a corrected five-protocol NS-3 audit, an expanded

seven-protocol CTP/RPL-MRHOF boundary sweep, and a stricter simulator matrix, keeping baseline ranking separate from stress evidence. Third, we use ablation, pooled trade-off statistics, and a corrected mechanism study to show that the observed gain comes mainly from Gateway support and is paid for through earlier node death.

II. RELATED WORK

Classical low-overhead WSN protocols still define the main engineering operating points. LEACH and HEED reduce long-range transmission burden through clustering, but their up-link quality is still sensitive to head placement and channel degradation [6], [11]. PEGASIS spreads load through chain forwarding but pays for that robustness with long routes and serialization [5]. TEEN reduces redundant transmissions under threshold-triggered sensing, but its reporting logic makes metric interpretation sensitive to how “expected” delivery is defined [7]. These protocols span the basic reliability-energy-latency trade-offs that any new design has to face.

Recent work has improved reliability through learning, hybrid control, or deployment adaptation [12]–[25]. AERIS is closer to this line of work than to purely energy-minimizing clustering, but it keeps the runtime logic deliberately simple: fixed coefficients, explicit control paths, and no training-time dependence. More recent 2024–2025 studies have also sharpened the surrounding context by surveying LLN path-finding requirements [26], SDN-style management over IoT/LLN deployments [27], and AI-driven energy-efficient routing in IoT-enabled WSNs [28].

Collection and RPL-style protocols are especially important for deployment realism. CTP builds a collection tree toward the sink using link/path quality, while RPL with MRHOF selects parents by minimizing path cost in a low-power and lossy-network DAG. Opportunistic variants such as ORPL/ORW go further by exploiting candidate forwarder sets and local link diversity. This paper does not claim to replace that whole family. Instead, it asks where a simpler fixed-rule WSN design still buys reliability, and where the stronger collection-tree family should remain the default.

III. PROTOCOL DESIGN

AERIS combines CAS for intra-cluster transmission mode selection, Gateway reinforcement for cluster-head to base-station uplinks, and Skeleton forwarding for sparse CH fall-

AERIS Round Workflow with Cascaded First-Hit Uplink Reinforcement

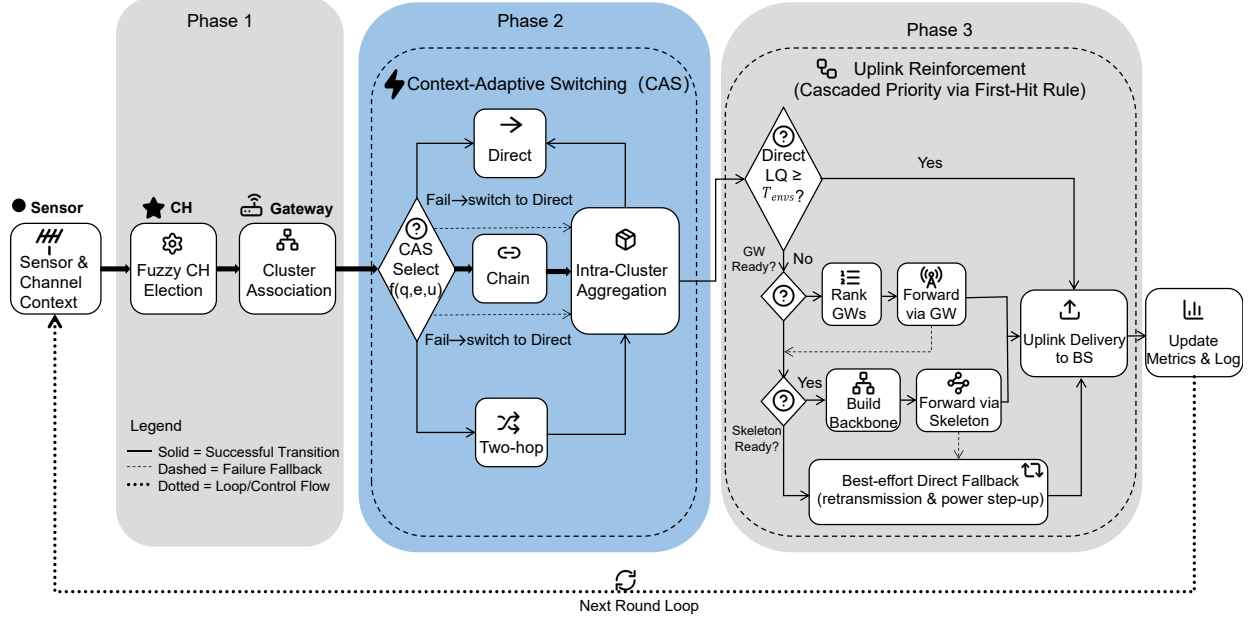


Fig. 1. AERIS round workflow with cascaded uplink reinforcement.

back routes. The goal is to add reliability support only where direct forwarding becomes unreliable, while keeping online logic deterministic and auditable.

Gateway candidates are ranked by a fixed linear score,

$$\text{Score}_i = \alpha \hat{E}_i + \beta \hat{C}_i + \gamma \hat{L}_i + \delta \hat{D}_i, \quad (1)$$

where $\hat{E}_i$, $\hat{C}_i$, $\hat{L}_i$, and $\hat{D}_i$ denote normalized residual energy, centrality, link quality, and CH-to-BS distance. In this implementation, centrality is a distance-aware neighborhood surrogate computed from local geometry, link quality is the current channel-estimated success probability for the candidate hop, and all scalar features are normalized to $[0, 1]$ before scoring. CAS uses another fixed linear rule over {direct, chain, two-hop}; the shared coefficients are held constant across environments in publication-tier runs, so mode selection is reproducible under identical inputs and seeds:

$$m^* = \arg \max_{m \in \{\text{direct}, \text{chain}, \text{two-hop}\}} (a_m \hat{q} + b_m \hat{e} + c_m \hat{u}), \quad (2)$$

where $\hat{q}$, $\hat{e}$, and $\hat{u}$ are normalized local channel quality, residual energy, and utilization terms, and the coefficients $\{a_m, b_m, c_m\}$ are fixed for each candidate mode. Skeleton forwarding is built as a sparse CH relay backbone after clustering and is refreshed round-by-round together with the CH set, but the corrected publication configuration later shows that it acts as a reserve fallback path rather than an actively exercised mechanism in the audited runs.

The forwarding logic is cascaded rather than multipath. A cluster head first checks direct uplink quality, then tries Gateway support if available, then Skeleton support if available, and finally makes one last direct attempt. In the strict-physics

regime, this “best-effort direct fallback” is a single direct transmission attempt without protocol-level retransmission.

In practice, the protocol only activates extra forwarding structure after a direct path has become unattractive under the current channel estimate. CAS decides whether a sensor-to-CH transfer stays direct, switches to a short chain, or uses a two-hop relay; Gateway support then targets the more fragile CH-to-BS segment; Skeleton remains a reserve fallback if Gateway support is unavailable. The control path is deliberately narrow: AERIS does not search a large policy space online, and the same inputs always trigger the same routing decisions.

AERIS is an engineering design rather than a formal routing theorem. It adds alternate forwarding structure only when the direct path looks weak, reacts to a small number of local conditions with fixed rules, and is therefore judged by how that choice changes delivery and cost in deployment.

A. Why the Modules Are Ordered This Way

The ordering of the modules is part of the design rather than an implementation detail. Direct transmission is checked first because it is the cheapest option whenever the channel still supports it. Gateway support comes next because the audited experiments show that the CH-to-BS segment is usually the first place where reliability collapses in harsher cells. Skeleton forwarding is deliberately placed behind Gateway support because it is intended to preserve a fallback path, not to replace the main uplink rule.

This ordering also keeps the policy interpretable. AERIS does not ask every round whether all possible forwarding structures should be explored. Instead, it asks a smaller sequence of questions: is direct transmission still reasonable, is there a nearby Gateway that can protect the uplink, and if not,

does the sparse Skeleton path exist? The protocol therefore behaves like a controlled escalation path rather than a broad route search.

B. Intended Operating Regime

The design is aimed at deployments where channel quality varies significantly across the field and where a modest increase in runtime complexity is acceptable if it stabilizes delivery. It is not aimed at the cleanest office-like regimes, where the longest-lived chain often remains the better choice, and it is not aimed at routing stacks that can afford heavier control traffic or training-time optimization. This intended operating regime is important in reading the evaluation, because it explains why the paper compares AERIS against classical baselines but interprets the result as deployment guidance rather than universal ranking.

IV. EVALUATION SETUP

The primary metric is packet delivery ratio over expected application-layer transmissions,

$$\text{PDR}_{\text{expected}} = \frac{N_{\text{delivered}}}{N_{\text{expected}}}, \quad (3)$$

where N_{expected} counts one packet per alive source node per round even if a protocol suppresses actual transmission. This denominator is kept fixed across protocols so that the reliability comparison is not quietly changed by protocol-specific reporting logic.

We use complementary settings. The first is a corrected five-protocol NS-3 audit against AERIS, LEACH, PEGASIS, HEED, and TEEN across four environments and 100, 500, and 1000 nodes ($n = 30$ per environment-node-protocol cell). The second is an expanded dual-machine NS-3 boundary sweep that adds CTP and RPL-MRHOF and covers 50, 100, 200, 300, 500, 800, and 1000 nodes. In the standalone NS-3 harness, CTP and RPL-MRHOF are implemented as collection-tree baselines: CTP favors progress-constrained link-cost forwarding, while RPL-MRHOF uses accumulated path cost. The third is a 3360-run NS-3 AERIS ablation matrix over the same four environments and seven node counts, comparing full AERIS with no-Gateway, no-CAS, and no-CH-score variants. The fourth setting is a strict-physics Python simulator matrix across four environments and 100–1000 nodes ($n = 3200$ per environment-node-protocol cell) with explicit collision and relay stress; in that layer, LEACH, HEED, and TEEN are reported in minimally adapted relay-enabled form for multi-hop comparability. To avoid relying on PDR alone, we also report total energy, lifetime, first-node death, and hop count in the frozen 100-node publication block. Mechanism and trade-off analysis use that block together with a corrected 400-replicate AERIS mechanism matrix.

The four environments are used consistently across all experiments: indoor office, indoor factory, outdoor suburban, and outdoor urban. They are meant to cover a progression from relatively benign links to much harsher settings. The node counts then separate small-network behavior from the

TABLE I
EXPERIMENTAL SETTINGS USED IN THE PAPER.

Layer	Platform	Protocols	Cells	Role
Classical audit	NS-3	5 protocols	4×3	repair anchor
Expanded audit	NS-3	7 protocols	4×7	boundary
Module ablation	NS-3	4 AERIS variants	4×7	attribution
Strict	Python	adapted 5-protocol	4×6	stress
Mechanism	Python	AERIS-only grid	4×3	attribution
Frozen block	pooled stats	5-protocol summary	100-node only	trade-off

denser cells where poor CH-to-BS links and relay pressure become more visible.

This role split is deliberate. The corrected five-protocol NS-3 audit asks whether AERIS beats the classical WSN baselines under the same independent-platform assumptions used in the original repair cycle. The expanded seven-protocol sweep asks whether that result survives contact with stronger collection-tree baselines. The NS-3 ablation asks which AERIS module actually carries the delivery gain, while the strict-physics simulator tests the classical-baseline story under stronger relay and collision assumptions in an adapted stress layer rather than in a second canonical ranking layer. Lifetime, first-node death, hop count, and total energy then show whether delivery gains come from spreading or concentrating load. Keeping these layers separate is one of the main repairs made after the earlier review cycle.

V. RESULTS

A. Expanded NS-3 Boundary

Figure 2 gives the main boundary result from the seven-protocol dual-machine NS-3 sweep. If the baseline family is restricted to classical WSN protocols, AERIS is rank-1 in 21 of 28 environment-node cells and top-2 in all 28. This confirms the original repair claim: AERIS is a strong reliability-first design relative to LEACH, PEGASIS, HEED, and TEEN outside the benign office regime.

The expanded audit also shows why that claim must be bounded. Once CTP and RPL-MRHOF are included, AERIS remains strongest mainly in outdoor suburban cells: it wins 6 of 7 scales and is essentially tied at 1000 nodes, where RPL-MRHOF leads by only 0.0001 PDR points in the dual sweep. In contrast, CTP leads the office cells, and RPL-MRHOF leads most factory and urban cells. AERIS is slightly ahead at factory-50, but falls 1.3–1.8 percentage points behind RPL-MRHOF at larger factory scales; in urban, the average gap to the winner is 5.69 points.

This is not a failed result. It is the actual deployment boundary. AERIS solves a weakness of the classical clustering/chain baselines by protecting fragile uplinks with Gateway support, but collection-tree and RPL-style baselines already solve part of that path-selection problem with richer parent choice. The strongest defensible claim is therefore not that AERIS is globally best, but that its rule-based control path delivers a large gain over classical WSN baselines and remains competitive or best in selected harsh-channel regimes, especially suburban cells.

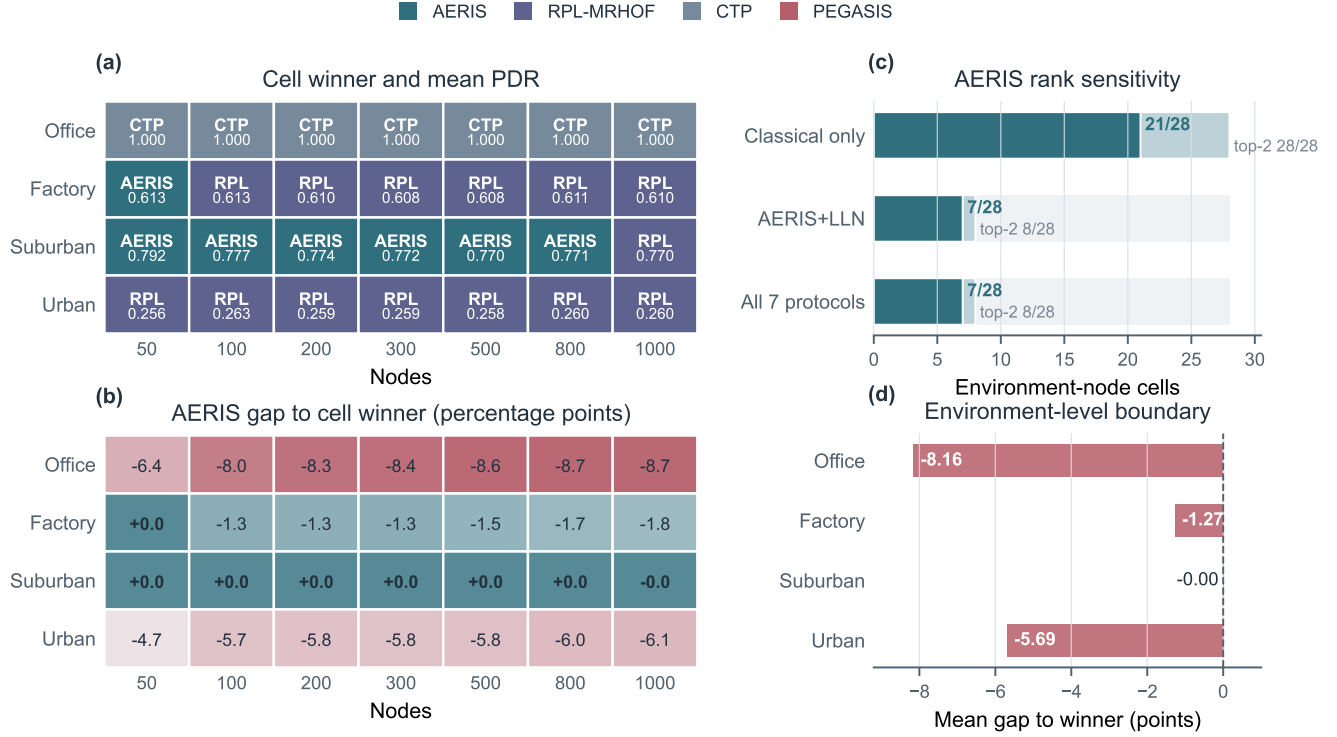


Fig. 2. Expanded seven-protocol NS-3 boundary map across four environments and 50–1000 nodes ($n = 30$ per cell). Panel (a) marks the winner in each cell. Panel (b) gives AERIS’s PDR gap to the cell winner in percentage points. Panels (c) and (d) show how AERIS’s rank changes when the baseline family expands from classical WSN protocols to CTP/RPL-MRHOF collection baselines.

B. Classical Audit and Stress Evidence

The corrected five-protocol NS-3 audit remains useful as the classical-baseline repair anchor. In that subset, PEGASIS is best in indoor office, while AERIS leads in factory, suburban, and urban conditions. At 1000 nodes, AERIS reaches 0.593 in indoor factory, 0.770 in outdoor suburban, and 0.200 in outdoor urban, compared with 0.324, 0.549, and 0.019 for PEGASIS. This is why the expanded result should be read as a boundary refinement rather than a reversal of the classical-baseline claim.

The strict-physics matrix asks whether the classical-baseline ordering survives once collision and relay stress are made explicit in the adapted layer. Figure 3 shows that it does. PEGASIS remains strongest in indoor office (0.991 at 100 nodes and 0.988 at 1000), whereas AERIS stays rank-1 in indoor factory, outdoor suburban, and outdoor urban. At 1000 nodes, AERIS remains well above PEGASIS in indoor factory (0.728 vs. 0.300) and outdoor suburban (0.728 vs. 0.473), and it still leads in outdoor urban (0.136 vs. 0.099).

This matrix is not the main fairness anchor because LEACH, HEED, and TEEN are minimally augmented with a simple relay layer for multi-hop comparability in this setting. We therefore do not interpret Fig. 3 as a second canonical protocol ranking. Its role is narrower: to show that AERIS’s harsh-channel advantage is not erased by explicit collision-and-relay stress even though absolute PDR falls for every protocol in the harder scales.

C. Ablation and Deployment Boundary

Figure 4 gives the expanded NS-3 module attribution test. It reruns full AERIS and three ablated variants over the four-environment, seven-scale matrix used in the boundary audit, for 3360 experiments in total. The result is sharper than the earlier frozen 100-node ablation: Gateway is the main gain carrier in factory and suburban cells. Removing Gateway costs 5.82 percentage points on average in factory and 7.58 points in suburban, with all seven node scales significant after Holm correction in both environments. It is essentially neutral in office and weak in urban, which is consistent with the boundary result: office does not need extra uplink protection, while urban links can become too poor for a nearby Gateway to rescue reliably.

CAS behaves differently. Removing CAS is beneficial in office and suburban cells, but hurts urban delivery by 0.91 points on average, with five of seven urban scales significant. This means CAS is not a universal gain module; it is useful mainly when a short local detour remains viable under harsh links. Removing the CH scoring component is near-neutral in all environments. This does not make the score irrelevant to the design, but it does show that the measured delivery gain in this publication configuration should not be attributed to head scoring.

The corrected 400-replicate mechanism matrix explains the same pattern inside AERIS. Gateway uplink PDR stays between 0.996 and 1.000 in office, factory, and suburban cells,

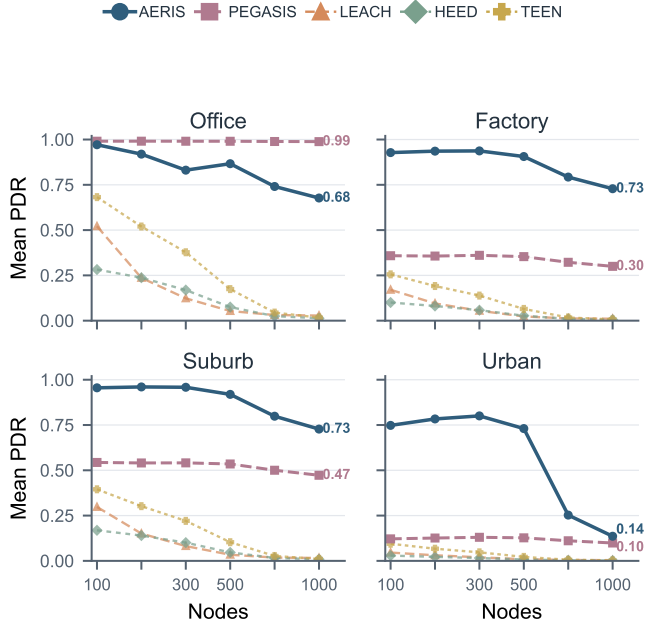


Fig. 3. Strict-physics Python adapted stress test across four environments and 100–1000 nodes ($n = 3200$ per cell). LEACH, HEED, and TEEN appear here in minimally relay-enabled form for strict-layer comparability; this layer is used as stress evidence, not as the sole fairness anchor.

but drops to 0.286 at urban-1000 exactly where end-to-end PDR collapses to 0.138. In that same cell, mean gateway uplink attempts fall to 63.3 per run, versus 136.9 in factory-1000 and 171.4 in suburban-1000, which is more consistent with the loss of viable reinforced uplinks than with a merely overloaded but still healthy gateway layer. A targeted follow-up rerun of those three harsh 1000-node cells on FatMachine reproduced the same means for PDR, gateway uplink PDR, FND, HND, gateway attempts, and two-hop usage to the reported precision, so this bottleneck pattern is not a one-off merge artifact. CAS is not uniformly helpful: two-hop share is small at 100 nodes, becomes dominant in several 500-node office/factory/suburban cells, and then falls again in urban-1000 as reliable alternatives disappear. Skeleton forwarding is not driving the observed gains in this publication configuration, because the corrected mechanism matrix records zero skeleton assignments across all 12 audited cells.

Taken together, the ablation and mechanism views say something concrete about how the protocol behaves. Gateway support matters most when the weak point of the path is the CH-to-BS hop and a viable reinforced uplink exists. CAS matters when a short local detour is still available. Skeleton is present in the control logic, but in the audited publication setting it is not the source of the measured gain.

The cost is just as clear. Table II shows that AERIS has the lowest total energy (131.1 J) but the shortest lifetime (215.5 rounds). PEGASIS uses slightly more energy (137.6 J) yet survives 300 rounds and delays first-node death to 70.7 rounds, compared with 15.8 for AERIS. Inside AERIS, first-node death falls from 13.7 rounds in office-100 to 1–

TABLE II
FROZEN 100-NODE PUBLICATION BLOCK POOLED ACROSS PROTOCOLS.
THE TABLE KEEPS PDR, ENERGY, LIFETIME, FND, AND HOP COUNT
VISIBLE IN ONE PLACE.

Method	PDR↑	Energy↓	Lifetime↑	FND↑	Hops↓
AERIS	0.674	131.1	215.5	15.8	1.98
PEGASIS	0.373	137.6	300.0	70.7	32.37
LEACH	0.260	165.4	300.0	0.0	1.71
HEED	0.414	165.2	300.0	11.4	2.00
TEEN	0.432	159.4	299.0	0.0	1.28

4.5 rounds in most 500/1000-node cells even when gateway uplinks remain strong. The degradation is not confined to the first failure. Half-node death drops from 114.9 rounds in office-100 to 34.1, 30.0, and 10.1 in the harsh 500-node cells, and to 4.5, 2.6, and 3.3 in the corresponding harsh 1000-node cells. Better delivery is bought with concentrated relay burden and rapidly propagating attrition once key forwarding roles are exhausted.

Figures 2–4 and Table II suggest a more concrete deployment rule. If the target cell behaves more like the office regime, or if the design target is closer to the pooled long-lifetime profile of PEGASIS (300 rounds lifetime, 70.7-round FND) than to maximum delivery, PEGASIS or another simpler chain-based solution is the safer default among classical WSN protocols. If the deployment can use richer collection-tree or RPL-style path selection, CTP/RPL-MRHOF should also be considered first in office, factory, and urban regimes. AERIS is easiest to justify when the application needs a simple auditable WSN control path, the baseline family is closer to classical clustering/chain protocols, or the cell resembles the suburban regime where AERIS remains best or effectively tied even after CTP/RPL-MRHOF are added.

D. When AERIS Should Not Be Chosen

The office regime is the clearest negative case for AERIS. When direct, chain, or collection-tree forwarding is already highly reliable, the extra AERIS control structure does not buy enough to offset earlier node death. PEGASIS remains a credible classical deployment option in benign settings, and CTP/RPL-MRHOF are stronger still in the expanded NS-3 audit.

The same caution applies when the operating goal is longevity rather than delivery, or when the system already has a mature LLN stack available. AERIS can look attractive because its total energy is low, but the audited statistics show that it pays for delivery by exhausting key forwarding roles earlier. For applications that tolerate some delivery loss but require long unattended operation, that trade-off can make PEGASIS or another simpler chain-based design the better choice. For applications that can absorb RPL-style control structure, the expanded audit shows that RPL-MRHOF can dominate AERIS in factory and urban cells.

E. How to Read the Boundary in Practice

The most useful way to read the results is not as a global leaderboard, but as a deployment filter with three explicit

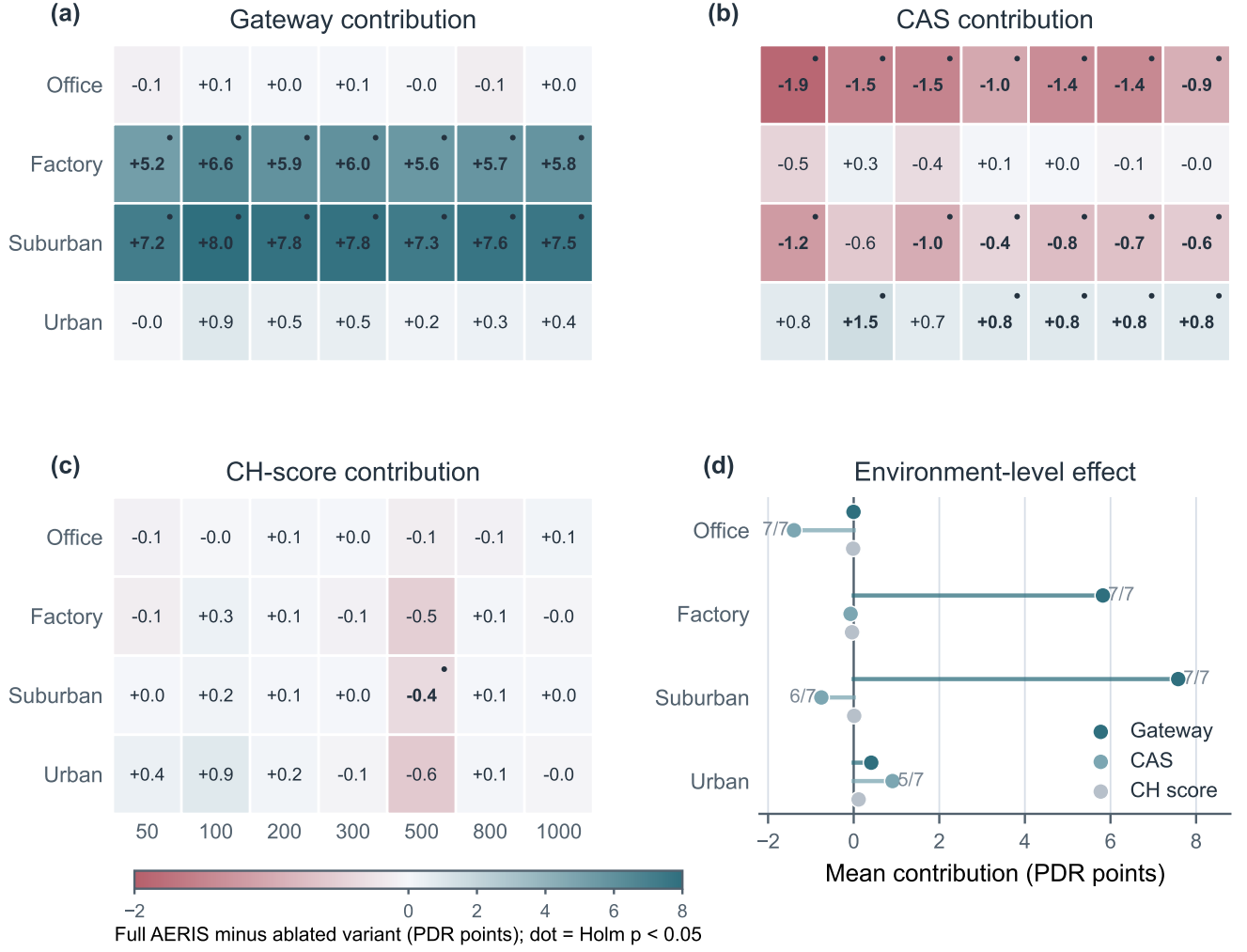


Fig. 4. Expanded NS-3 AERIS ablation across four environments and 50–1000 nodes ($n = 30$ per cell). Heatmaps show the PDR contribution of each module, computed as full AERIS minus the ablated variant in percentage points; dots mark cells significant after Holm correction. Gateway carries the factory/suburban gain, CAS is environment-dependent, and CH-score removal is near-neutral in this audited configuration.

checks. First, is a collection-tree or RPL-style stack available and acceptable? If yes, Fig. 2 says it should be considered seriously in office, factory, and urban cells. Second, if the deployment is constrained to lightweight classical WSN routing, is PEGASIS-like delivery in the harsh cells acceptable for the application? If yes, the simpler chain-based option remains attractive. Third, is AERIS-like early node attrition acceptable? If the application cannot tolerate first-node death near 1–4.5 rounds in harsh 500/1000-node cells, then the added reliability is not worth the cost even when AERIS wins on PDR.

If the answer to the first two questions is no and the third is yes, the evidence favors AERIS. If, instead, the cell looks closer to the office regime, or if RPL-MRHOF/CTP can be deployed, the same logic leads to the opposite recommendation. The office and urban cases therefore act as built-in negative controls for the paper’s main claim.

The remaining protocols help bracket this interpretation. LEACH and HEED show what happens when clustered forwarding is kept simple but the uplink is left more exposed to

poor channels. TEEN reminds the reader that event-triggered reporting complicates PDR interpretation, which is why the pooled block keeps energy, lifetime, FND, and hops visible next to delivery. CTP and RPL-MRHOF then show how much of AERIS’s advantage is really a classical-baseline gap rather than a general advantage over collection-tree routing.

VI. DISCUSSION AND LIMITATIONS

The main message is simple. AERIS is strongest as a simple reliability-first alternative to classical WSN baselines, especially in the suburban harsh-channel regime; it is not the default choice for benign office-like regimes, lifetime-dominated deployments, or settings where collection-tree/RPL-style routing is available and acceptable.

That reading is narrower than a universal performance claim, but it is also more useful. The paper now tells the reader where AERIS should be used, where it should not, and why the answer changes across baseline families and environments. For an engineering routing paper, that is a stronger contribution

than reporting another average improvement number without a boundary.

Several limits remain. The expanded NS-3 audit includes CTP and RPL-MRHOF collection baselines, but it is still not the full space of low-power routing: opportunistic routing, ORPL/ORW-style candidate forwarding, coding-based reliability, channel-hopping systems, and full standards-compliant RPL stacks remain outside the current evidence package. The strict-physics layer is simulator based and should be read as deployment-stress evidence rather than external ground truth; it also uses minimally adapted relay-enabled versions of LEACH, HEED, and TEEN for strict-layer comparability, so it is not a second native-baseline benchmark. Its MAC abstraction still omits full 802.15.4/TSCH/CSMA-CA details such as ACK loops, backoff dynamics, capture effects, and channel hopping. Recent LLN work likewise continues to emphasize that scalable path management and control-plane flexibility remain open problems in low-power IoT deployments [26], [27]. Because TEEN is event-triggered, PDR_{expected} should be read together with energy, lifetime, FND, and hop count rather than on its own. The corrected mechanism study identifies Gateway as the dominant harsh-channel module and CAS as environment dependent, but it also shows that Skeleton forwarding is dormant in the audited publication configuration (`skeleton_assignments = 0` in all 12 audited cells); here it should be read as a reserve design element rather than a gain-bearing mechanism. The current study is also limited to static topologies, the tested environment taxonomy, and non-public review-time artifacts. Recent routing work on enhanced clustering, hotspot mitigation, adaptive DRL policies, and learned HEED/LEACH variants [28]–[33] suggests several directions for future comparison, but those are outside the present repair-focused submission.

For the settings covered here, the guidance is simple: use AERIS when a lightweight rule-based WSN design needs stronger delivery than classical baselines provide and the application can tolerate the observed early-node attrition envelope; prefer PEGASIS or a simpler chain-based solution when benign conditions and longevity dominate; prefer CTP/RPL-style collection routing when that stack is available and the environment resembles the office, factory, or urban cells in the expanded audit. The paper is not arguing that fixed-rule routing always beats richer low-power networking stacks, nor that Gateway support always beats path-cost parent selection. The audited claim is narrower: under these conditions, a simple rule-based design can buy a meaningful delivery advantage over classical baselines, and the cost and boundary of that choice are visible rather than hidden.

A. Evidence Provenance

Because this paper is built around a corrected resubmission package, provenance must be explicit. The classical-baseline repair statement is tied to the corrected five-protocol NS-3 rerun, while the seven-protocol dual-machine sweep is used to map the stronger CTP/RPL-MRHOF boundary. The strict-physics simulator stresses the classical-baseline story under

stronger assumptions, and the AERIS-only mechanism grid is used to attribute the observed gain to Gateway support and selected two-hop use. The pooled 100-node block then provides a compact cross-protocol view of energy, lifetime, FND, and hops. As a consistency check, we also reran the three harsh 1000-node mechanism cells (factory, suburban, urban) in a separate targeted FatMachine batch and recovered the same means for the key mechanism metrics to the reported precision; this follow-up confirms the published bottleneck pattern but does not replace the full 12-cell audited matrix. This separation is intentional: earlier drafts blurred independent-platform evidence and stress evidence too easily. Review-time artifact access is still limited, but the internal provenance chain is now explicit and consistent.

B. Implications for Low-Power Routing

One broader implication of these results is that the vulnerable part of the path matters more than the nominal route structure. In the audited experiments, the decisive difference against classical baselines is whether the weak CH-to-BS segment is explicitly protected when the environment becomes harsher. AERIS does this through Gateway support, whereas the classical baselines leave that burden mostly to their default forwarding structure. CTP and RPL-MRHOF reduce the same weakness in a different way: they build a collection path with stronger parent choice. That is why they narrow or erase AERIS’s advantage in several environments.

The results also argue for a more explicit treatment of inactive modules in protocol papers. The corrected mechanism audit shows that Skeleton forwarding is present in the design but dormant in the publication configuration, so it should be read as reserve logic rather than as the source of the measured gain.

Finally, the trade-off shown here is a useful reminder that low energy and long lifetime are not interchangeable goals. AERIS spends little total energy in the pooled 100-node block, yet it still loses on lifetime because the burden is concentrated on the nodes that remain effective under bad channels. For practical system design, this means that “energy-efficient” should not be treated as a proxy for “long-lived” once routing assistance becomes asymmetric across the network.

VII. CONCLUSION

This paper revisited AERIS as a reliable WSN routing design and reevaluated it with a corrected five-protocol NS-3 audit, an expanded seven-protocol CTP/RPL-MRHOF boundary sweep, a large strict-physics simulator matrix, and a corrected mechanism study. Across these layers, the picture is sharper than a simple leaderboard: AERIS is a strong reliability-first alternative to classical WSN baselines, but its advantage is narrowed by richer collection-tree routing.

The more important result is the boundary behind the ranking. In the audited experiments, AERIS’s delivery gain over classical baselines comes mainly from protecting the uplink through Gateway support, while the cost appears as earlier first-node death and shorter lifetime. When CTP and

RPL-MRHOF are added, AERIS remains strongest mainly in the suburban regime and loses in benign office, most factory, and urban cells. In that sense, AERIS should be read less as a universal routing replacement than as a practical choice for deployments where a simple auditable control path is required and classical WSN baselines leave too much delivery loss.

The next step is not to make the claim broader rhetorically, but to make the boundary sharper empirically. That means comparing against full RPL/ORPL/ORW-style stacks, adding more realistic MAC behavior, and testing whether Gateway-style reinforcement can be combined with collection-tree parent selection without reproducing the early-node-death cost.

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